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Research Paper

The F-regularity of algebraic sets related to commutator matrices

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ABSTRACT

We study the algebraic sets of pairs of matrices defined by the vanishing of the anti-diagonal as well as the cross-diagonal of their commutator matrix. We prove that, over a field of prime characteristic, the coordinate ring of the latter is always F-regular and, with exactly one exception, so is that of the former, thus proving a conjecture of Kadyrsizova and Yerlanov.

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1. Introduction

Let $X = (x_{ij})_{1\leq i,j\leq n}$ and $Y = (y_{ij})_{1\leq i,j\leq n}$ be $n \times n$ matrices of indeterminates over a field $\Bbbk$. The *commutator matrix* of X and Y is $C := XY - YX = (c_{ij})_{1\leq i,j\leq n}$. Let $\mathfrak{d}$ and $\mathfrak{a}$ denote the ideals generated by the diagonal and the anti-diagonal entries of C, respectively. Then $\mathfrak{c} := \mathfrak{d} + \mathfrak{a}$ is the ideal generated by the *cross-diagonal* of C.

Algebraic sets defined by commutator matrices have been studied extensively since at least the 1950s. Such algebraic sets were proved to be irreducible by Motzkin and Taussky [15], Gerstenhaber [3], and Guralnick [5] under varying assumptions. Hochster

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(1984) conjectured that these algebraic sets are reduced. It is straightforward to verify this when $n \leq 2$. Soon after, Thompson [18] settled the conjecture in the case $n = 3$; the conjecture remains open for $n \geq 4$. Knutson [13] introduced the algebraic set of *nearly commuting matrices*, i.e., the set defined by the vanishing of all off-diagonal entries of C, and conjectured that it is a reduced complete intersection, with two irreducible components, one of which is the variety of commuting $n \times n$ matrices. This was proved by Young [19] who also studied the algebraic set of matrices with zero diagonal commutator, i.e., the set defined by the vanishing of $\mathfrak{d}$, and showed that it is an irreducible complete intersection.

Since the introduction of tight closure [8], it has been a question of considerable interest to determine the F-singularities, i.e., the tight closure properties, of various algebraic sets. The classic examples that have been studied extensively are determinantal, symmetric determinantal, and Pfaffian rings, which are all known to be F-regular, see for example [10, Theorem 7.14]. How about coordinate rings defined by the aforementioned algebraic sets related to commutator matrices? Partial answers were recently obtained by Kadyrsizova [11] and Kadyrsizova and Yerlanov [12]:

Theorem 1.1. *Let $\Bbbk$ be a field of positive characteristic, X and Y be $n \times n$ matrices of indeterminates, $C = (c_{ij})_{1 \leq i,j \leq n}$ their commutator matrix, and $\mathfrak{d}$, $\mathfrak{a}$, and $\mathfrak{c}$ are as defined above. Then:*

(1) *[11, Theorem 6] The ring $\Bbbk[X,Y]/(c_{ij})_{1 \leq i,j \leq n,\ i \neq j}$ is F-pure in the cases $n \leq 3$.*
(2) *[12, page 30] The ring $\Bbbk[X,Y]/(c_{ij})_{1 \leq i,j \leq n,\ i \neq j}$ is not F-pure when $n = 4$ and $\operatorname{char}(\Bbbk) = 2$.*
(3) *[12, Theorem 2.3] The ring $\Bbbk[X,Y]/\mathfrak{d}$ is F-regular.*
(4) *[12, Theorems 3.7, 3.9, 4.3] The rings $\Bbbk[X,Y]/\mathfrak{c}$ and $\Bbbk[X,Y]/\mathfrak{a}$ are F-pure.*

In this paper, we determine precisely when the rings $\Bbbk[X,Y]/\mathfrak{c}$ and $\Bbbk[X,Y]/\mathfrak{a}$ are F-regular, thus proving [12, Conjectures 5.1, 5.2]. Our main result is:

Theorem 1.2. *Let $\Bbbk$ be a field of positive characteristic, X and Y be $n \times n$ matrices of indeterminates, and $\mathfrak{c}$ and $\mathfrak{a}$ denote the ideals generated by cross-diagonal and anti-diagonal entries of the commutator matrix $XY - YX$, respectively. Then $\Bbbk[X,Y]/\mathfrak{c}$ is F-regular for each n, while $\Bbbk[X,Y]/\mathfrak{a}$ is F-regular exactly when $n \neq 2$.*

Using [17, Theorem 4.3], one immediately obtains the characteristic-zero version:

Corollary 1.3. *Let $\Bbbk$ be a field of characteristic zero, X and Y be $n \times n$ matrices of indeterminates, and $\mathfrak{c}$ and $\mathfrak{a}$ denote the ideals generated by cross-diagonal and anti-diagonal entries of the commutator matrix $XY - YX$, respectively. Then $\Bbbk[X,Y]/\mathfrak{c}$ has rational singularities for each n, while $\Bbbk[X,Y]/\mathfrak{a}$ has rational singularities exactly when $n \neq 2$.*

Corollary 1.4. *In any characteristic, the ideal* $\mathfrak{c}$ *is prime for each* n*, while the ideal* $\mathfrak{a}$ *is prime exactly when* $n \neq 2$*.*

2. Preliminaries

We recall the relevant definitions. Let R be a Noetherian commutative ring of prime characteristic $p > 0$ and R° denote the complement of all minimal primes of R. The ring R is *strongly F-regular* if for each $c \in R^\circ$, there exists $q = p^e$ such that the R-linear map $R \to R^{1/q},\ 1 \mapsto c^{1/q}$ splits. There are different notions of F-regularity, namely weak F-regularity, F-regularity, and strong F-regularity. For an $\mathbb{N}$-graded ring, these coincide, so we do not make a distinction, and will only use *F-regular*; the equivalence was proved by Lyubeznik and Smith [14, Corollaries 4.3, 4.4] if the ring is *F-finite*, i.e., the embedding $R \to R^{1/p}$ is module-finite, and by Hochster [6, page 237] in general. From now on, unless otherwise stated, we will assume that R is $\mathbb{N}$-graded.

A weaker condition than F-regularity is F-purity. A ring R is *F-pure* if the embedding $R \to R^{1/p}$ is pure, i.e., the induced map

$$R \otimes M \to R^{1/p} \otimes M$$

is injective for any R-module M.

Although these two properties do not deform in general ([16, Theorem 1.1] and [1, Example 4.8]), they do deform for Gorenstein rings:

Theorem 2.1. *([9, Theorem 4.2], [1, Theorem 3.4]) Let* f *be an* R*-regular element. If* R *is Gorenstein and* R/fR *is* F*-regular (respectively,* F*-pure), then* R *is* F*-regular (respectively,* F*-pure).*

Useful ways to check F-regularity and F-purity are Glassbrenner's and Fedder's criteria, recorded below in the graded case. For an ideal I and a prime power $q = p^e$, let $I^{[q]}$ denote the ideal generated by the qth powers of the elements in I.

Theorem 2.2. *(Glassbrenner's criterion, [4, Theorem 3.1]) Let* S *be a polynomial ring over an* F*-finite field of prime characteristic* $p > 0$ *and* $\mathfrak{m}$ *denote its homogeneous maximal ideal. If* I *is a homogeneous ideal, then* S/I *is* F*-regular if and only if for each* $c \in (S/I)^\circ$*, we have* $c(I^{[q]} :_S I) \nsubseteq \mathfrak{m}^{[q]}$ *for some* $q = p^e$*.*

Theorem 2.3. *(Fedder's criterion, [1, Theorem 1.12]) Let* S *be a polynomial ring over an* F*-finite field of prime characteristic* $p > 0$ *and* $\mathfrak{m}$ *denote its homogeneous maximal ideal. If* I *is a homogeneous ideal, then* S/I *is* F*-pure if and only if* $(I^{[p]} :_S I) \nsubseteq \mathfrak{m}^{[p]}$*.*

In the special case when I is generated by a regular sequence, we have

$$(I^{[q]} :_S I) = (\omega^{q-1}) + I^{[q]},$$

where ω denotes the product of minimal generators of I. Therefore, to show that I in this case defines an F-pure (respectively, F-regular ring), it is necessary and sufficient to show that $\omega^{p-1} \notin \mathfrak{m}^{[p]}$ (respectively, that for each $c \in (S/I)^\circ$, we have $c\omega^{q-1} \notin \mathfrak{m}^{[q]}$ for some $q = p^e$). We remark that all the rings that we will prove to be F-pure or F-regular in this paper are complete intersections.

It turns out that one does not need to check the definition of F-regularity of a ring R and in turn, the condition in Glassbrenner's criterion, for every $c \in R^\circ$. It suffices to verify it for just one certain nonzero $c \in R^\circ$, called a *test element*. There are many test elements. One way to obtain a test element is from the singular locus of the ring:

Theorem 2.4. *([7, Theorem 3.4]) Let R be an F-finite reduced ring. Then every element $c \in R^\circ$ such that R_c is regular has a power which is a test element.*

Theorem 2.5. *([2, Proposition 2.5]) Let R be an F-finite and F-pure ring. Then the ideal generated by all test elements is radical.*

Combining these, one obtains:

Corollary 2.6. *Let R be an F-finite and F-pure ring. Then every element $c \in R^\circ$ such that R_c is regular is a test element.*

3. Proof of Theorem 1.2

Let $\Bbbk$ be a field of prime characteristic $p > 0$. To ease the notations, set $A_n := \Bbbk[X, Y]/\mathfrak{c}$ and $B_n := \Bbbk[X, Y]/\mathfrak{a}$ where X and Y are $n \times n$ matrices of indeterminates, and their commutator matrix C, and $\mathfrak{a}$ and $\mathfrak{c}$ are as defined earlier. To show the F-regularity of a ring, it suffices to do so for a faithfully flat extension [8, Proposition 4.12]. Therefore, for the rest of the paper, we will assume $\Bbbk$ is algebraically closed.

Our strategy is to use deformation techniques and simplify the task at hand by decomposing $\Bbbk[X, Y]/\mathfrak{c}$ into a tensor product of rings of smaller dimensions. Towards this, we will start by proving that some rings of small dimensions are F-regular.

Proposition 3.1. *Let $W = (w_{ij})$ and $Z = (z_{ij})$ be 3×3 matrices of indeterminates. Then the ring*

$$T = \Bbbk[W, Z]/(f_1, f_2, f_3, f_4)$$

is F-regular, where

$$\begin{aligned} f_1 &= w_{21}z_{12} - w_{12}z_{21} + w_{23}z_{32} - w_{32}z_{23}, \\ f_2 &= w_{22}z_{21} - w_{21}z_{22} + w_{23}z_{31} - w_{31}z_{23} + w_{21}z_{11} - w_{11}z_{21}, \\ f_3 &= w_{13}z_{32} - w_{32}z_{13} + w_{12}z_{22} - w_{22}z_{12} + w_{11}z_{12} - w_{12}z_{11}, \end{aligned}$$

$$f_4 = w_{12}z_{21} - w_{21}z_{12} + w_{13}z_{31} - w_{31}z_{13}.$$

Proof. We first claim that the set of elements

$$\begin{gathered} w_{11},\ w_{21},\ w_{23},\ w_{32},\ w_{33},\ z_{11},\ z_{13},\ z_{22},\ z_{32},\ z_{33}, \\ w_{12} - z_{21},\ w_{13} - z_{31},\ w_{22} - z_{12},\ w_{31} - z_{23}, \\ f_1,\ f_2,\ f_3,\ f_4 \end{gathered} \tag{3.1}$$

is a homogeneous system of parameters for $\Bbbk[W, Z]$, and thus a regular sequence on $\Bbbk[W, Z]$. Indeed, this set has 18 elements, and $\dim \Bbbk[W, Z] = 18$, hence it suffices to show that the ring $\Bbbk[W, Z]$ modulo the ideal they generate has Krull dimension 0. This ring is isomorphic to

$$\frac{\Bbbk[w_{12},\ w_{13},\ w_{22},\ w_{31}]}{(w_{12}^2,\ \ w_{22}w_{12} - w_{31}^2,\ \ w_{22}^2,\ \ w_{13}^2)},$$

which is evidently of dimension 0. In particular, the ring T is a complete intersection.

Next we show that T is F-pure. Since F-purity deforms for complete intersections by Theorem 2.1, it suffices to show that the homomorphic image of T modulo the ideal generated by

$$w_{11},\ w_{21},\ w_{23},\ w_{32},\ w_{33},\ z_{11},\ z_{13},\ z_{22},\ z_{32},\ z_{33},$$

a subset of (3.1), is F-pure. Indeed, this homomorphic image is the complete intersection:

$$\frac{\Bbbk[w_{12},\ w_{13},\ w_{22},\ w_{31},\ z_{12},\ z_{21},\ z_{23},\ z_{31}]}{(w_{12}z_{21},\ \ -w_{22}z_{21} + w_{31}z_{23},\ \ w_{22}z_{12},\ \ w_{13}z_{31})},$$

which is evidently F-pure by Fedder's criterion since

$$\big((w_{12}z_{21})(-w_{22}z_{21} + w_{31}z_{23})(w_{22}z_{12})(w_{13}z_{31})\big)^{p-1}$$

equals $(w_{12}w_{13}w_{22}w_{31}z_{12}z_{21}z_{23}z_{31})^{p-1}$ modulo $\mathfrak{m}^{[p]}$.

We turn to F-regularity and first need a test element. We compute a maximal minor of the Jacobian matrix of the ideal defining T:

$$\det \frac{\partial(f_1, f_2, f_3, f_4)}{\partial(w_{31}, w_{32}, z_{31}, z_{32})} = \det \begin{pmatrix} 0 & -z_{23} & 0 & -z_{13} \\ -z_{23} & 0 & -z_{13} & 0 \\ 0 & w_{23} & 0 & w_{13} \\ w_{23} & 0 & w_{13} & 0 \end{pmatrix} = (w_{23}z_{13} - w_{13}z_{23})^2.$$

Hence, by Corollary 2.6, the polynomial $f \coloneqq w_{23}z_{13} - w_{13}z_{23}$ is a test element once we show that it is in T°. Observe that the homomorphic image of $\Bbbk[W, Z]$ modulo the ideal generated by the $18 = \dim \Bbbk[W, Z]$ elements

$$\begin{gathered} w_{11},\ w_{12},\ w_{31},\ w_{33},\ z_{11},\ z_{21},\ z_{32},\ z_{33}, \\ w_{13}-z_{31},\ w_{21}-z_{22},\ w_{23}-z_{13},\ w_{32}-z_{23},\ w_{13}+w_{22}+z_{12}, \\ f,\ f_1,\ f_2,\ f_3,\ f_4 \end{gathered} \tag{3.2}$$

is isomorphic to

$$\Bbbk[w_{13},\ w_{21},\ w_{22},\ w_{23},\ w_{32}]/(g,\ g_1,\ g_2,\ g_3,\ g_4) \tag{3.3}$$

where

$$\begin{aligned} g &= w_{23}^2 - w_{13}w_{32}, \\ g_1 &= w_{32}^2 + w_{13}w_{21} + w_{21}w_{22}, \\ g_2 &= w_{21}^2 - w_{13}w_{23}, \\ g_3 &= w_{22}^2 + w_{13}w_{22} - w_{23}w_{32}, \\ g_4 &= w_{13}^2 + w_{13}w_{21} + w_{21}w_{22}. \end{aligned}$$

The radical of the above defining ideal contains w_{21} since

$$\begin{aligned} w_{21}^4 = (w_{21}w_{23} + w_{23}^2 - w_{13}w_{32})g_1 + (w_{21}^2 - w_{22}w_{23})g_2 + (-w_{23}^2 + w_{13}w_{32} + w_{21}w_{32})g_3 \\ + (-w_{23}^2 + w_{13}w_{32} - w_{22}w_{32})g_4 + (w_{13}^2 + w_{22}^2 - w_{23}w_{32} - w_{32}^2)g, \end{aligned}$$

and by examining the defining polynomials, one can show that it also contains w_{13}, w_{22}, w_{23} and w_{32}. The ring (3.3) is thus of dimension 0, i.e., (3.2) forms a homogeneous system of parameters for $\Bbbk[W,Z]$, hence a regular sequence. In particular, f is T-regular.

Now we show that T is F-regular. By Glassbrenner's criterion, it suffices to show that

$$(f)(f_1f_2f_3f_4)^{p-1} \notin \mathfrak{m}^{[p]}$$

where $\mathfrak{m}$ denotes the homogeneous maximal ideal of $\Bbbk[W,Z]$. This is equivalent to verifying that the image of $(f)(f_1f_2f_3f_4)^{p-1}$ is nonzero in $\Bbbk[W,Z]/\mathfrak{m}^{[p]}$. We will in fact, prove something stronger. For the remainder of the proof, all elements denote their homomorphic images in

$$\Bbbk[W,Z]/\big((w_{11},w_{21},w_{31},w_{33},z_{11},z_{12},z_{32},z_{33}) + \mathfrak{m}^{[p]}\big). \tag{3.4}$$

We will show that

$$(w_{23}z_{13})^{p-2}(f)(f_1f_2f_3f_4)^{p-1} \neq 0.$$

For degree reasons, this element is a scalar multiple of $(w_{12}w_{13}w_{22}w_{23}w_{32}z_{13}z_{21}z_{22}z_{23}z_{31})^{p-1}$, hence it suffices to find the corresponding coefficient. Note that in the ring (3.4) one has

$$\begin{aligned} f &= w_{23}z_{13} - w_{13}z_{23}, \\ f_1 &= -w_{12}z_{21} - w_{32}z_{23}, \\ f_2 &= w_{22}z_{21} + w_{23}z_{31}, \\ f_3 &= -w_{32}z_{13} + w_{12}z_{22}, \\ f_4 &= w_{12}z_{21} + w_{13}z_{31}. \end{aligned}$$

Since w_{22} and z_{22} only appear in f_2 and f_3, respectively, to obtain $w_{22}^{p-1}z_{22}^{p-1}$, the only monomial terms that matter in f_2^{p-1} and f_3^{p-1} are ones divisible by w_{22}^{p-1} and z_{22}^{p-1}, respectively. Thus

$$\begin{aligned} &(w_{23}z_{13})^{p-2}(f)(f_1f_2f_3f_4)^{p-1} \\ &= (w_{23}z_{13})^{p-2}(f)(f_1^{p-1})\big(w_{22}z_{21}\big)^{p-1}\big(w_{12}z_{22}\big)^{p-1}(f_4^{p-1}) \\ &= (w_{12}w_{22}z_{21}z_{22})^{p-1}(w_{23}z_{13})^{p-2}(f)(f_1^{p-1})(f_4^{p-1}). \end{aligned}$$

Using similar arguments for f_1^{p-1} and f_4^{p-1}, the above element is simplified to

$$\begin{aligned} &(w_{12}w_{22}z_{21}z_{22})^{p-1}(w_{23}z_{13})^{p-2}(f)(-w_{12}z_{21} - w_{32}z_{23})^{p-1}(w_{12}z_{21} + w_{13}z_{31})^{p-1} \\ &= (w_{12}w_{22}z_{21}z_{22})^{p-1}(w_{23}z_{13})^{p-2}(f)\big(w_{32}z_{23}\big)^{p-1}\big(w_{13}z_{31}\big)^{p-1} \\ &= (w_{12}w_{13}w_{22}w_{32}z_{21}z_{22}z_{23}z_{31})^{p-1}(w_{23}z_{13})^{p-2}(f). \end{aligned}$$

Substituting f, the above equals

$$\begin{aligned} &(w_{12}w_{13}w_{22}w_{32}z_{21}z_{22}z_{23}z_{31})^{p-1}(w_{23}z_{13})^{p-2}(w_{23}z_{13} - w_{13}z_{23}) \\ &= (w_{12}w_{13}w_{22}w_{23}w_{32}z_{13}z_{21}z_{22}z_{23}z_{31})^{p-1}, \end{aligned}$$

which is nonzero in the ring (3.4), as claimed. □

Next we will prove the F-regularity of A_n for some small values of n. Since the trace of the commutator matrix $C = XY - YX$ is 0, we can choose the generators of $\mathfrak{c}$ to be the first $n-1$ entries of the diagonal counting from the upper-left corner, together with the anti-diagonal entries of C. We remark that the number of minimal generators of $\mathfrak{c}$ depends on the parity of n, since the diagonal and anti-diagonal of an $n \times n$ matrix share a common entry if and only if n is odd.

Proposition 3.2. *The ring $A_3 = \Bbbk[X,Y]/\mathfrak{c}$, where X and Y are 3×3 matrices of indeterminates and $\mathfrak{c}$ denotes the ideal generated by the cross-diagonal of their commutator matrix, is F-regular.*

Proof. The ideal $\mathfrak{c}$ is generated by the elements $c_{11}, c_{22}, c_{31}, c_{13}$, where

$$\begin{aligned}
c_{11} &= x_{12}y_{21} - x_{21}y_{12} + x_{13}y_{31} - x_{31}y_{13},\\
c_{22} &= x_{21}y_{12} - x_{12}y_{21} + x_{23}y_{32} - x_{32}y_{23},\\
c_{31} &= x_{32}y_{21} - x_{21}y_{32} + x_{31}y_{11} - x_{11}y_{31} - x_{31}y_{33} + x_{33}y_{31},\\
c_{13} &= x_{12}y_{23} - x_{23}y_{12} + x_{11}y_{13} - x_{13}y_{11} + x_{13}y_{33} - x_{33}y_{13}.
\end{aligned}$$

We observe that

$$f \coloneqq x_{13}y_{23}(x_{13}y_{31} - x_{31}y_{13}) = -\det \frac{\partial(c_{11}, c_{22}, c_{31}, c_{13})}{\partial(x_{11}, x_{32}, y_{11}, y_{31})}$$

is a maximal minor of the Jacobian matrix of the ideal defining A_3, and hence by Corollary 2.6 and the fact that A_3 is F-pure (Theorem 1.1. (4)), f is a test element once we show that it is in A_3°. Indeed, the homomorphic image of $\Bbbk[X, Y]$ modulo the ideal generated by the $18 = \dim \Bbbk[X, Y]$ elements

$$\begin{gathered}
x_{12},\ x_{22},\ x_{31},\ x_{33},\ y_{11},\ y_{22},\ y_{33},\\
x_{11} - y_{13},\ x_{13} - y_{23},\ x_{13} - y_{31},\ x_{21} - y_{12},\ x_{23} - y_{32},\ x_{32} - y_{21},\\
c_{11},\ c_{22},\ c_{31},\ c_{13},\ f
\end{gathered} \tag{3.5}$$

is isomorphic to

$$\frac{\Bbbk[x_{11},\ x_{13},\ x_{21},\ x_{23},\ x_{32}]}{(x_{13}^2 - x_{21}^2,\ \ x_{21}^2 + x_{23}^2 - x_{13}x_{32},\ \ -x_{11}x_{13} - x_{21}x_{23} + x_{32}^2,\ \ x_{11}^2 - x_{21}x_{23},\ \ x_{13}^4)}$$

which is evidently of dimension 0, i.e., (3.5) forms a homogeneous system of parameters for $\Bbbk[X, Y]$, hence a regular sequence. In particular, f is A_3-regular.

By Glassbrenner's criterion, to prove the F-regularity of A_3, it suffices to show that

$$(f)(c_{11}c_{22}c_{31}c_{13})^{p^2-1} \notin \mathfrak{m}^{[p^2]} \tag{3.6}$$

where $\mathfrak{m}$ denotes the homogeneous maximal ideal of $\Bbbk[X, Y]$. We will prove something stronger. For the remainder of the proof, all elements denote their homomorphic images in

$$\Bbbk[X, Y]/\big((x_{12}, x_{22}, x_{31}, x_{33}, y_{11}, y_{22}, y_{33}) + \mathfrak{m}^{[p^2]}\big). \tag{3.7}$$

We will show that the element

$$(x_{13}^{p^2-3})(y_{23}y_{31})^{p^2-2}(f)(c_{11}c_{22}c_{31}c_{13})^{p^2-1}$$

is nonzero in the ring (3.7). For degree reasons, this element equals a scalar multiple of $(x_{11}x_{13}x_{21}x_{23}x_{32}y_{12}y_{13}y_{21}y_{23}y_{31}y_{32})^{p^2-1}$. In the ring (3.7) one has

$$\begin{aligned} f &= x_{13}^2 y_{23} y_{31}, \\ c_{11} &= -x_{21}y_{12} + x_{13}y_{31}, \\ c_{22} &= x_{21}y_{12} + x_{23}y_{32} - x_{32}y_{23}, \\ c_{31} &= x_{32}y_{21} - x_{21}y_{32} - x_{11}y_{31}, \\ c_{13} &= -x_{23}y_{12} + x_{11}y_{13}. \end{aligned}$$

We first substitute f:

$$\begin{aligned} &(x_{13}^{p^2-3})(y_{23}y_{31})^{p^2-2}(f)(c_{11}c_{22}c_{31}c_{13})^{p^2-1} \\ &= (x_{13}y_{23}y_{31})^{p^2-1}(c_{11}c_{22}c_{31}c_{13})^{p^2-1} \end{aligned}$$

Since we already have the factor $(x_{13}y_{23}y_{31})^{p^2-1}$, any additional monomial term that contains one of x_{13}, y_{23}, and y_{31} may be ignored. Substituting c_{11}, c_{22}, c_{31}, and c_{13} with this observation, one has

$$\begin{aligned} &(x_{13}y_{23}y_{31})^{p^2-1}\big(-x_{21}y_{12}\big)^{p^2-1}(x_{21}y_{12}+x_{23}y_{32})^{p^2-1}(x_{32}y_{21}-x_{21}y_{32})^{p^2-1} \\ &\qquad\qquad (-x_{23}y_{12}+x_{11}y_{13})^{p^2-1} \\ &= (x_{13}x_{21}y_{12}y_{23}y_{31})^{p^2-1}(x_{21}y_{12}+x_{23}y_{32})^{p^2-1}(x_{32}y_{21}-x_{21}y_{32})^{p^2-1} \\ &\qquad\qquad (-x_{23}y_{12}+x_{11}y_{13})^{p^2-1} \end{aligned}.$$

We have a new monomial factor $(x_{21}y_{12})^{p^2-1}$. Using similar arguments, we obtain

$$\begin{aligned} &(x_{13}x_{21}y_{12}y_{23}y_{31})^{p^2-1}\big(x_{23}y_{32}\big)^{p^2-1}\big(x_{32}y_{21}\big)^{p^2-1}\big(x_{11}y_{13}\big)^{p^2-1} \\ &= (x_{11}x_{13}x_{21}x_{23}x_{32}y_{12}y_{13}y_{21}y_{23}y_{31}y_{32})^{p^2-1} \end{aligned}$$

which is nonzero in the ring (3.7), as claimed. □

Proposition 3.3. *The ring $A_4 = \Bbbk[X,Y]/\mathfrak{c}$, where X and Y are 4×4 matrices of indeterminates and $\mathfrak{c}$ denotes the ideal generated by the cross-diagonal of their commutator matrix, is F-regular.*

Proof. The ideal $\mathfrak{c}$ is generated by $\{c_{11}, c_{22}, c_{33}, c_{41}, c_{32}, c_{23}, c_{14}\}$ where

$$\begin{aligned} c_{11} &= x_{12}y_{21} - x_{21}y_{12} + x_{13}y_{31} - x_{31}y_{13} + x_{14}y_{41} - x_{41}y_{14}, \\ c_{22} &= x_{21}y_{12} - x_{12}y_{21} + x_{23}y_{32} - x_{32}y_{23} + x_{24}y_{42} - x_{42}y_{24}, \\ c_{33} &= x_{31}y_{13} - x_{13}y_{31} - x_{23}y_{32} + x_{32}y_{23} + x_{34}y_{43} - x_{43}y_{34}, \end{aligned}$$

$$c_{41} = x_{41}y_{11} - x_{11}y_{41} - x_{21}y_{42} + x_{42}y_{21} - x_{31}y_{43} + x_{43}y_{31} - x_{41}y_{44} + x_{44}y_{41},$$
$$c_{32} = x_{31}y_{12} - x_{12}y_{31} - x_{22}y_{32} + x_{32}y_{22} - x_{32}y_{33} + x_{33}y_{32} + x_{34}y_{42} - x_{42}y_{34},$$
$$c_{23} = x_{21}y_{13} - x_{13}y_{21} + x_{22}y_{23} - x_{23}y_{22} + x_{23}y_{33} - x_{33}y_{23} + x_{24}y_{43} - x_{43}y_{24},$$
$$c_{14} = x_{11}y_{14} - x_{14}y_{11} + x_{12}y_{24} - x_{24}y_{12} + x_{13}y_{34} - x_{34}y_{13} + x_{14}y_{44} - x_{44}y_{14}.$$

We observe that

$$f \coloneqq x_{12}x_{13}y_{24}(x_{14}y_{41} - x_{41}y_{14})(x_{23}y_{32} - x_{32}y_{23}) = -\det \frac{\partial(c_{11}, c_{22}, c_{33}, c_{41}, c_{32}, c_{23}, c_{14})}{\partial(x_{11}, x_{22}, x_{42}, y_{11}, y_{21}, y_{22}, y_{31})}$$

is a maximal minor of the Jacobian matrix of the ideal defining A_4, and thus by Corollary 2.6 and the fact that A_4 is F-pure (Theorem 1.1. (4)), f is a test element once we show that it is in A_4°. Observe that the homomorphic image of $\Bbbk[X, Y]$ modulo the ideal generated by the $32 = \dim \Bbbk[X, Y]$ elements

$$\begin{gathered} x_{31},\ x_{32},\ x_{33},\ x_{41},\ x_{44},\ y_{11},\ y_{13},\ y_{21},\ y_{22},\ y_{33},\ y_{44}, \\ x_{12} - x_{13},\ x_{12} - x_{14},\ x_{12} - x_{23}, \\ x_{11} - y_{14},\ x_{12} - y_{24},\ x_{12} - y_{32},\ x_{12} - y_{41}, \\ x_{21} - y_{12},\ x_{22} - y_{23},\ x_{24} - y_{42}, \\ x_{34} - y_{43},\ x_{42} - y_{34},\ x_{43} - y_{31}, \\ c_{11},\ c_{22},\ c_{33},\ c_{41},\ c_{32},\ c_{23},\ c_{14},\ f \end{gathered} \tag{3.8}$$

is isomorphic to

$$\frac{\Bbbk[x_{11},\ x_{12},\ x_{21},\ x_{22},\ x_{24},\ x_{34},\ x_{42},\ x_{43}]}{\begin{gathered}(x_{12}^2 - x_{21}^2 + x_{12}x_{43},\ \ x_{12}^2 + x_{21}^2 + x_{24}^2 - x_{12}x_{42},\ \ -x_{12}^2 + x_{34}^2 - x_{12}x_{43} - x_{42}x_{43}, \\ -x_{11}x_{12} - x_{21}x_{24} + x_{43}^2,\ \ -x_{12}x_{22} + x_{24}x_{34} - x_{42}^2 - x_{12}x_{43}, \\ x_{22}^2 + x_{24}x_{34} - x_{12}x_{43},\ \ x_{11}^2 + x_{12}^2 - x_{21}x_{24} + x_{12}x_{42},\ \ x_{12}^7)\end{gathered}}$$

which is evidently of dimension 0, i.e., the elements (3.8) form a homogeneous system of parameters for $\Bbbk[X, Y]$, hence a regular sequence. In particular, f is A_4-regular. By Glassbrenner's criterion, to prove the F-regularity of A_4, it suffices to show that

$$(f)(c_{11}c_{22}c_{33}c_{41}c_{32}c_{23}c_{14})^{p-1} \notin \mathfrak{m}^{[p]}$$

where $\mathfrak{m}$ denotes the homogeneous maximal ideal of $\Bbbk[X, Y]$. We will prove something stronger, that the element

$$(x_{12}x_{13}x_{32}x_{41}y_{14}y_{23}y_{24})^{p-2}(f)(c_{11}c_{22}c_{33}c_{41}c_{32}c_{23}c_{14})^{p-1} \tag{3.9}$$

is nonzero, where all elements, for the remainder of the proof, denote their homomorphic images in the ring

$$\Bbbk[X,Y]/\big((x_{11},x_{22},x_{24},x_{33},x_{44},y_{31},y_{32},y_{33},y_{41},y_{43},y_{44})+\mathfrak{m}^{[p]}\big). \tag{3.10}$$

We replace the polynomials with their corresponding homomorphic images:

$$\begin{aligned}
f &= x_{12}x_{13}x_{32}x_{41}y_{14}y_{23}y_{24},\\
c_{11} &= x_{12}y_{21}-x_{21}y_{12}-x_{31}y_{13}-x_{41}y_{14},\\
c_{22} &= x_{21}y_{12}-x_{12}y_{21}-x_{32}y_{23}-x_{42}y_{24},\\
c_{33} &= x_{31}y_{13}+x_{32}y_{23}-x_{43}y_{34},\\
c_{41} &= x_{41}y_{11}-x_{21}y_{42}+x_{42}y_{21},\\
c_{32} &= x_{31}y_{12}+x_{32}y_{22}+x_{34}y_{42}-x_{42}y_{34},\\
c_{23} &= x_{21}y_{13}-x_{13}y_{21}-x_{23}y_{22}-x_{43}y_{24},\\
c_{14} &= -x_{14}y_{11}+x_{12}y_{24}+x_{13}y_{34}-x_{34}y_{13}.
\end{aligned}$$

For degree reasons, the element (3.9) equals a scalar multiple of

$$(x_{12}x_{13}x_{14}x_{21}x_{23}x_{31}x_{32}x_{34}x_{41}x_{42}x_{43}y_{11}y_{12}y_{13}y_{14}y_{21}y_{22}y_{23}y_{24}y_{34}y_{42})^{p-1}.$$

We first substitute f in the element (3.9):

$$\begin{aligned}
&(x_{12}x_{13}x_{32}x_{41}y_{14}y_{23}y_{24})^{p-2}(f)(c_{11}c_{22}c_{33}c_{41}c_{32}c_{23}c_{14})^{p-1}\\
&=(x_{12}x_{13}x_{32}x_{41}y_{14}y_{23}y_{24})^{p-1}(c_{11}c_{22}c_{33}c_{41}c_{32}c_{23}c_{14})^{p-1}.
\end{aligned}$$

Since we already have the factor $(x_{12}x_{13}x_{32}x_{41}y_{14}y_{23}y_{24})^{p-1}$, any additional monomial term that contains one of these variables may be ignored. Substituting c_{22} with this observation, we obtain

$$\begin{aligned}
&(x_{12}x_{13}x_{32}x_{41}y_{14}y_{23}y_{24})^{p-1}\big(c_{11}(x_{21}y_{12})c_{33}c_{41}c_{32}c_{23}c_{14}\big)^{p-1}\\
&=(x_{12}x_{13}x_{21}x_{32}x_{41}y_{12}y_{14}y_{23}y_{24})^{p-1}(c_{11}c_{33}c_{41}c_{32}c_{23}c_{14})^{p-1}
\end{aligned}$$

We have a new monomial factor $(x_{21}y_{12})^{p-1}$. Using similar arguments for c_{11}, c_{41}, and c_{23}, the above element equals

$$\begin{aligned}
&(x_{12}x_{13}x_{21}x_{32}x_{41}y_{12}y_{14}y_{23}y_{24})^{p-1}\big((-x_{31}y_{13})c_{33}(x_{42}y_{21})c_{32}(-x_{23}y_{22})c_{14}\big)^{p-1}\\
&=(x_{12}x_{13}x_{21}x_{23}x_{31}x_{32}x_{41}x_{42}y_{12}y_{13}y_{14}y_{21}y_{22}y_{23}y_{24})^{p-1}\big(c_{33}c_{32}c_{14}\big)^{p-1}
\end{aligned}$$

With the new monomial factor $(x_{23}x_{31}x_{42}y_{13}y_{21}y_{22})^{p-1}$, we simplify c_{33}, c_{32}, and c_{14}:

$$(x_{12}x_{13}x_{21}x_{23}x_{31}x_{32}x_{41}x_{42}y_{12}y_{13}y_{14}y_{21}y_{22}y_{23}y_{24})^{p-1}\big((-x_{43}y_{34})(x_{34}y_{42})(-x_{14}y_{11})\big)^{p-1}$$
$$= (x_{12}x_{13}x_{14}x_{21}x_{23}x_{31}x_{32}x_{34}x_{41}x_{42}x_{43}y_{11}y_{12}y_{13}y_{14}y_{21}y_{22}y_{23}y_{24}y_{34}y_{42})^{p-1},$$

which is nonzero in the ring (3.10), as claimed. □

Proof of Theorem 1.2. Recall that $A_n = \Bbbk[X,Y]/\mathfrak{c}$, where X and Y are $n \times n$ matrices of indeterminates and $\mathfrak{c}$ denotes the ideal generated by the cross-diagonal of their commutator matrix. The rings A_1 and A_2 are F-regular [11, pages 201-202]. We now assume $n \geq 3$. It is known that A_n is an F-pure complete intersection [12, Theorem 3.4, Theorem 3.7, Theorem 3.9]. In particular, we have

$$\dim\ A_{2k-1} = 8k^2 - 12k + 6$$

and

$$\dim\ A_{2k} = 8k^2 - 4k + 1$$

where $k \geq 2$. We will show that the rings A_n are F-regular by induction on n. We already proved the base cases that A_3 and A_4 are F-regular in Propositions 3.2 and 3.3. Assume that A_m is F-regular for any $m < n$. We will prove that A_n is F-regular. If $n = 2k+1$ for some $k \geq 2$, set

$$\Omega_1 \coloneqq \{x_{ij}, y_{ij} \mid (i,j) = (k+1, l) \text{ or } (l, k+1) \text{ where } 1 \leq l < n\}$$

We have

$$\frac{A_n}{(\Omega_1)} \cong \frac{\Bbbk[X',Y']}{\mathfrak{c}(X'Y' - Y'X')} \otimes_{\Bbbk} \frac{\Bbbk\begin{bmatrix} x_{n,k+1} & x_{k+1,n} \\ y_{n,k+1} & y_{k+1,n} \end{bmatrix}}{(x_{n,k+1}y_{k+1,n} - x_{k+1,n}y_{n,k+1})}$$
$$\cong A_{n-1} \otimes_{\Bbbk} \frac{\Bbbk[w_1,\ w_2,\ w_3,\ w_4]}{(w_1w_4 - w_2w_3)},$$

where X' and Y' are obtained from X and Y, respectively, by deleting the $(k+1)^{\text{th}}$ row and column, and w_1, w_2, w_3, w_4 are new indeterminates. Since determinantal rings are F-regular [10, Theorem 7.14], the ring on the right-hand side is F-regular due to [9, Theorem 7.45] and the induction hypothesis, and thus so is $A_n/(\Omega_1)$. To show that A_n is F-regular, it suffices to show that Ω_1 forms a regular sequence on A_n by Theorem 2.1. Since A_n is a complete intersection, it suffices to show that the dimension of A_n drops by exactly $|\Omega_1|$ after taking modulo by the ideal generated by Ω_1. Indeed, we have

$$\dim\ A_n/(\Omega_1) + |\Omega_1| = \dim\ A_{n-1} + \dim \frac{\Bbbk[w_1,\ w_2,\ w_3,\ w_4]}{(w_1w_4 - w_2w_3)} + 2(4k-1)$$
$$= \dim\ A_{2k} + 3 + 2(4k-1)$$

$$\begin{aligned}
&= (8k^2 - 4k + 1) + 3 + 2(4k - 1)\\
&= 8k^2 + 4k + 2\\
&= 8(k+1)^2 - 12(k+1) + 6\\
&= \dim\ A_{2k+1}\\
&= \dim\ A_n.
\end{aligned}$$

Now suppose $n = 2k$ for some $k \geq 3$. Set

$$\Omega_2 \coloneqq \{x_{ij}, y_{ij} \mid (i,j) = (k,l), (k+1,l), (l,k) \text{ or } (l,k+1) \text{ where } 1 \leq l < k \text{ or } k+1 < l < n\}.$$

We have

$$\frac{A_n}{(\Omega_2)} \cong \frac{\Bbbk[X'', Y'']}{\mathfrak{c}(X''Y'' - Y''X'')} \otimes_{\Bbbk} T' \cong\ A_{n-2} \otimes_{\Bbbk} T'$$

where X'' and Y'' are obtained from X and Y, respectively, by deleting the kth and $(k+1)^{\text{th}}$ rows and columns and

$$T' \coloneqq \frac{\Bbbk \begin{bmatrix} x_{k,k} & x_{k,k+1} & x_{k,n} \\ x_{k+1,k} & x_{k+1,k+1} & x_{k+1,n} \\ x_{n,k} & x_{n,k+1} & 0 \end{bmatrix} \begin{bmatrix} y_{k,k} & y_{k,k+1} & y_{k,n} \\ y_{k+1,k} & y_{k+1,k+1} & y_{k+1,n} \\ y_{n,k} & y_{n,k+1} & 0 \end{bmatrix}}{\begin{pmatrix} x_{k+1,k}y_{k,k+1} - x_{k,k+1}y_{k+1,k} + x_{k+1,n}y_{n,k+1} - x_{n,k+1}y_{k+1,n}, \\ x_{k+1,k+1}y_{k+1,k} - x_{k+1,k}y_{k+1,k+1} + x_{k+1,n}y_{n,k} - x_{n,k}y_{k+1,n} + x_{k+1,k}y_{k,k} - x_{k,k}y_{k+1,k}, \\ x_{k,n}y_{n,k+1} - x_{n,k+1}y_{k,n} + x_{k,k+1}y_{k+1,k+1} - x_{k+1,k+1}y_{k,k+1} + x_{k,k}y_{k,k+1} - x_{k,k+1}y_{k,k}, \\ x_{k,k+1}y_{k+1,k} - x_{k+1,k}y_{k,k+1} + x_{k,n}y_{n,k} - x_{n,k}y_{k,n} \end{pmatrix}}.$$

Since $T'[w_{33}, z_{33}] = T$ where T denotes the ring in Proposition 3.1, the ring T' is a direct summand of T, an F-regular ring. Thus T' is F-regular by [8, Proposition 4.12] and hence so is $A_n/(\Omega_2)$ due to [9, Theorem 7.45] and the induction hypothesis. We claim that the elements in Ω_2 form a regular sequence on A_n and will show it using similar arguments to the previous case. Indeed, since T' is a complete intersection (because T is, and T is exactly T' with two more unrelated variables), we have

$$\dim T' = 16 - 4 = 12.$$

Now we verify the identity we need to claim that Ω_2 is indeed a regular sequence:

$$\begin{aligned}
\dim\ A_n/(\Omega_2) + |\Omega_2| &= \dim\ A_{n-2} + \dim T' + 2(8k - 12)\\
&= \dim\ A_{2k-2} + \dim T' + 2(8k - 12)\\
&= (8(k-1)^2 - 4(k-1) + 1) + 12 + 2(8k - 12)
\end{aligned}$$

$$\begin{aligned} &= 8k^2 - 4k + 1 \\ &= \dim\ A_{2k} \\ &= \dim\ A_n. \end{aligned}$$

Hence the claim holds, which implies that A_n is F-regular by Theorem 2.1.

Regarding $B_n = \Bbbk[X, Y]/\mathfrak{a}$, where X and Y are $n \times n$ matrices of indeterminates, it is known that B_1 is F-regular and B_2 is not even a domain, let alone F-regular [11, page 202]. For $n \geq 3$, since A_n is a complete intersection [12, Theorem 3.4], it is isomorphic to B_n modulo a regular sequence. Since A_n is F-regular, so is B_n for any $n \geq 3$ by Theorem 2.1. □

Data availability

No data was used for the research described in the article.

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